Session 03 (AIML) - Assignment Question

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
#session 3
#Q1

# Functions help us reuse code and avoid writing the same code again.

def welcome():
    print("Welcome to Python Programming!")

welcome()
```

OUTPUT:

```
Welcome to Python Programming!
```

Q2.

PROGRAM:

```
#Q2

def greet(name):
    print("Hello,", name + "! Welcome back.")

greet("Shruti")
greet("sairaj")
```

OUTPUT:

```
Hello, Shruti! Welcome back.
Hello, sairaj! Welcome back.
```

Q3.

PROGRAM:

```python
#Q3

# Default parameters allow a function to work even when no argument is passed.

def show_message(message="Hello"):
    print(message)

show_message()
show_message("Good Morning")
```

C

```
Hello
Good Morning
```

Q4.

PROGRAM:

```python
#Q4

def calculate_sum(a, b):
    return a + b

result = calculate_sum(10, 20)

print("Sum =", result)
```

OUTPUT:

```
Sum = 30
```

PROGRAM:

Difference between print() and return:

```
#Q4
#Difference between print() and return:

def print_sum(a, b):
    print("Sum =", a + b)

print_sum(5, 3)

def calculate_sum(a, b):
    return a + b

value = calculate_sum(5, 3)
print("Returned Value =", value)
```

OUTPUT:

```
Sum = 8
Returned Value = 8
```

Q5.

PROGRAM:

```
#Q5

# Positional arguments: Values are passed in order.
# Keyword arguments: Parameter names are specified.

def student_info(name, age):
    print("Name:", name)
    print("Age:", age)

# Positional arguments
student_info("Shruti", 18)
z
# Keyword arguments
student_info(age=18, name="Shruti")
```

OUTPUT:

```
Name: Shruti
Age: 18
Name: Shruti
Age: 18
```


Q6.

PROGRAM:

```
#Q6

num = int(input("Enter a number: "))
def square(num):
    return num * num
print("Square =", square(num))

def cube(num):
    return num * num * num
print("Cube =", cube(num))
```

OUTPUT:

```
Enter a number:  25
Square = 625
Cube = 15625
```

Q7.

PROGRAM:

```
#Q7

def countdown(n):
    if n == 0:       # Base case
        return
    print(n)
    countdown(n - 1)

countdown(10)
```

OUTPUT:

```
10
9
8
7
6
5
4
3
2
1
```

Q8.

PROGRAM:

```python
#Q8

def factorial(n):
    # Base case
    if n == 1 or n == 0:
        return 1

    return n * factorial(n - 1)

num = int(input("Enter a number: "))
print("Factorial =", factorial(num))
```

OUTPUT:

```
Enter a number:  25
Factorial = 15511210043330985984000000
```

Q9.

PROGRAM:

```python
#Q9
# Global variable
total = 0

def add_value(x):
    global total  # Access and modify the global variable
    total += x
    local_var = x  # Local variable

    print("Total =", total)

# Call the function multiple times
add_value(10)
add_value(20)
add_value(30)

# Trying to access a local variable outside the function
# This will cause an error because local_var exists only inside add_value()
print(local_var)
```

OUTPUT:

```
Total = 10
Total = 30
Total = 60
```

Fix Code

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[17], line 19
     15 add_value(30)
     17 # Trying to access a local variable outside the function
     18 # This will cause an error because local_var exists only inside add_value()
---> 19 print(local_var)

NameError: name 'local_var' is not defined
```

Q10.

PROGRAM:

```python
#Q10

def get_number():
    return int(input("Enter a number (0 to exit): "))

def is_even(num):
    return num % 2 == 0

def power(base, exp=2):
    return base ** exp

def show_result(num):
    if is_even(num):
        print(num, "is Even")
    else:
        print(num, "is Odd")

    print("Square =", power(num))

while True:
    number = get_number()

    if number == 0:
        print("Program Ended")
        break

    show_result(number)
    print()
```

OUTPUT:

```
Enter a number (0 to exit):  5
5 is Odd
Square = 25

Enter a number (0 to exit):  0
Program Ended
```